

AI & Innovation Hackathon Challenges

Problem Statement 1: Real-time Audio Deepfake Neutralizer

Develop a low-latency edge audio processing model that detects synthetic voice cloning artifacts during active phone calls and injects warning watermark tones to prevent social engineering fraud.

Problem Statement 2: Autonomous Swarm Pathfinding for Search & Rescue

Design a decentralized reinforcement learning agent for autonomous terrestrial rovers navigating rubble in collapsed buildings to locate survivors using thermal and carbon dioxide sensors.

Problem Statement 3: Low-Power Neuromorphic Vision Sensor for Wildlife Tracking

Create an event-camera vision pipeline that processes microsecond optical spikes on solar-powered camera traps, classifying endangered feline species while consuming under 50 milliwatts.